Supplementary Material for "Learned Direct Actuation versus Tuned and RL-Tuned Pure Pursuit: A Controlled Comparison for Path Tracking"

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This supplement contains material moved from the main paper for length: the data split and the front end from image to reference (Figs. S1–S2), the waypoint spacing of every circuit reference (Table S1), the results on the synthetic test contours (Table S2), the per-circuit results of the naive-protocol contrast (Table S3), and further views of the test-circuit results (Figs. S3–S5). All numbers are generated from the released result files.

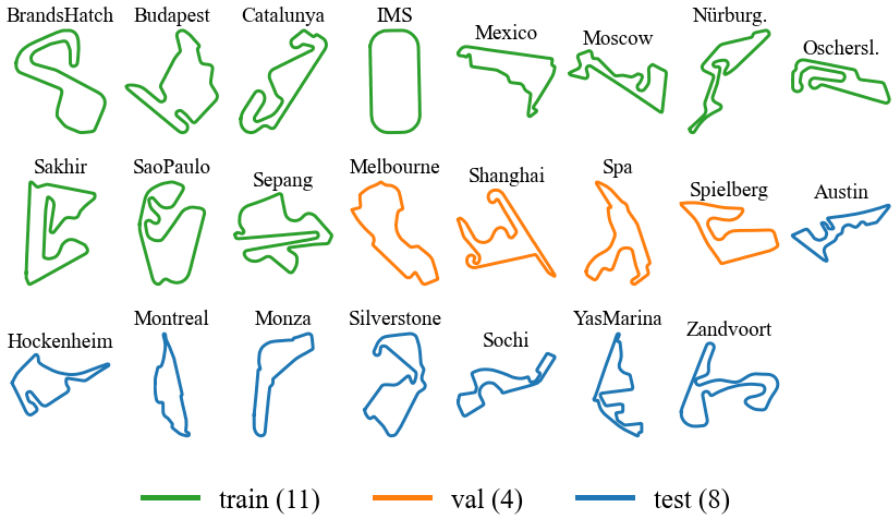


Figure S1: The 23 extracted F1TENTH references, with color indicating the split. Besides the three circuits that trained the naive-protocol bead (Yas Marina, Montreal, Hockenheim), five more form the test set; none of them is used to train, tune or select any controller.

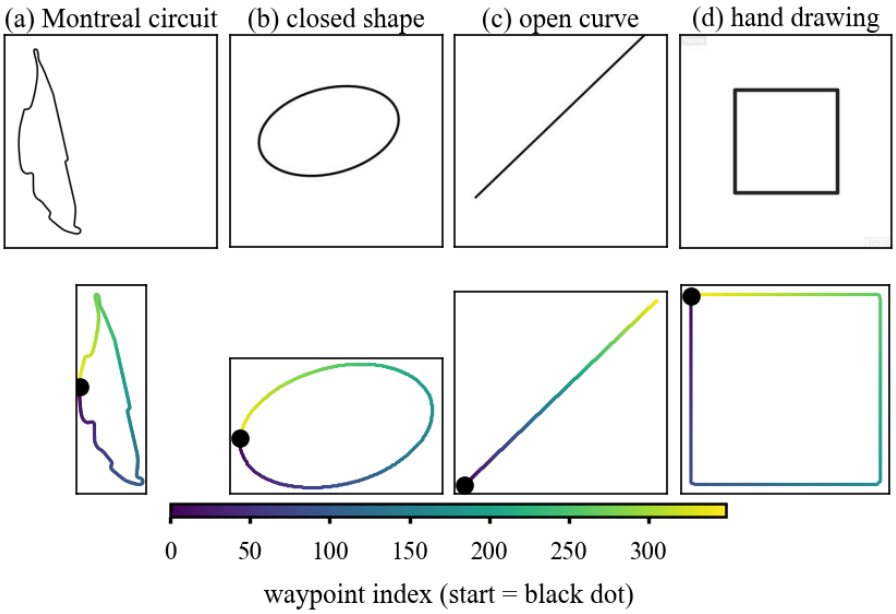


Figure S2: From image to reference path. Upper row: four inputs, namely one test circuit rendered as an image, one closed and one open synthetic test contour, and a hand drawing. Lower row: the ordered paths obtained from them, shaded by waypoint number, with waypoint 0 shown as a black dot.

Table S1: Waypoint spacing of the extracted circuit references (median and range of the distance between consecutive waypoints). The expert of Eq. (3) of the paper aims at the next waypoint, so its target lies 11.1–23.5 cm ahead depending on the circuit; Pure Pursuit interpolates its lookahead point along the polyline.

Circuit	Split	Waypoints	Median spacing [cm]	Range [cm]
BrandsHatch	train	350	20.1	19.6–21.6
Budapest	train	350	17.7	16.7–18.5
Catalunya	train	350	19.5	18.3–20.9
IMS	train	350	13.3	12.3–13.3
MexicoCity	train	349	14.6	13.3–15.4
MoscowRaceway	train	350	16.5	15.9–17.5
Nuerburgring	train	350	16.5	16.1–17.5
Oschersleben	train	350	17.4	17.0–18.4
Sakhir	train	350	21.9	20.3–23.0
SaoPaulo	train	350	19.9	19.7–21.3
Sepang	train	350	23.5	21.8–24.6
Melbourne	val	350	14.7	12.9–15.5
Shanghai	val	350	21.7	20.3–23.1
Spa	val	350	16.6	15.0–17.8
Spielberg	val	350	16.6	5.6‡–17.7
Austin	test	350	14.6	10.5‡–15.6
Hockenheim	test	350	15.1	14.1–16.1
Montreal	test	350	11.1	10.7–11.8
Monza	test	350	13.0	12.2–13.6
Silverstone	test	350	16.6	16.1–17.6
Sochi	test	350	15.4	14.3–16.5
YasMarina	test	350	15.1	13.3–16.1
Zandvoort	test	350	21.5	20.4–23.0

‡Shortest segment is the last one, where the extracted loop stops short of its start and its final points hook back slightly; it is not a junction between pieces (no circuit reference has a break): Spielberg 5.6 cm, loop open by 8.2 cm, minimum of the other segments 15.7 cm; Austin 10.5 cm, loop open by 4.6 cm, minimum of the other segments 12.9 cm.

Table S2: Accuracy on the synthetic test set of 100 contours; Geom@0.10 is given as average ± standard deviation over the contours, and the best entry of each set is bold. *, †: naive protocol; §: Fixed PP law on the bead.

Set / Controller	Geom@0.10 [%]	RMSE [cm]	Ordered [%]	Time [s]
Closed shapes (50)				
Geometric expert (3)	99.1 ± 0.3	1.12	99.1	24.4
Bead-PP (PP law on bead)§	99.1 ± 0.4	0.69	99.1	24.8
Bead-DAgger	99.1 ± 0.4	1.17	99.1	24.4
Bead-DAgger→PPO	99.1 ± 0.4	1.56	99.1	24.4
Bead-PPO (scratch)	98.2 ± 1.1	2.94	99.1	22.0
Fixed PP (tuned)	99.1 ± 0.3	0.63	99.1	10.8
RL–PP (joint L_d, g) [4]	98.8 ± 0.5	1.46	99.1	10.8
Bead, naive protocol*	96.2 ± 2.8	3.90	99.0	25.1
RL–PP, naive protocol†	97.8 ± 1.4	2.35	99.0	10.7
Open curves (50)				
Geometric expert (3)	98.6 ± 0.1	1.06	99.1	15.5
Bead-PP (PP law on bead)§	98.6 ± 0.1	0.71	99.1	15.7
Bead-DAgger	98.6 ± 0.1	1.09	99.1	15.5
Bead-DAgger→PPO	98.6 ± 0.1	1.46	99.1	15.4
Bead-PPO (scratch)	98.4 ± 0.7	2.42	99.1	13.8
Fixed PP (tuned)	98.7 ± 0.1	0.69	99.1	8.5
RL–PP (joint L_d, g) [4]	98.7 ± 0.1	1.19	99.1	8.5
Bead, naive protocol*	98.4 ± 0.6	2.58	99.1	15.9
RL–PP, naive protocol†	98.5 ± 0.4	1.65	99.1	8.5

Table S3: Scores on the three circuits that trained the naive-protocol bead, as Geom@0.10 [%] and RMSE [cm]; bold marks the best entry per column.

	Hockenheim		Montreal		Yas Marina	
Controller	G@0.10	RMSE	G@0.10	RMSE	G@0.10	RMSE
Geometric expert (3)	99.1	0.94	99.3	1.24	99.1	1.36
Bead-DAgger	99.1	0.87	99.3	1.35	99.1	1.40
Bead, naive protocol*	99.1	2.45	96.1	3.45	95.6	4.05
Fixed PP (hand-set gains)	95.7	3.62	90.8	5.54	89.8	5.17
Fixed PP (tuned)	99.1	0.27	99.3	0.41	99.1	0.50
RL–PP (joint L_d, g) [4]	98.7	1.64	99.0	1.60	95.5	2.40
RL–PP, naive protocol†	96.1	3.72	94.2	4.40	91.9	4.58

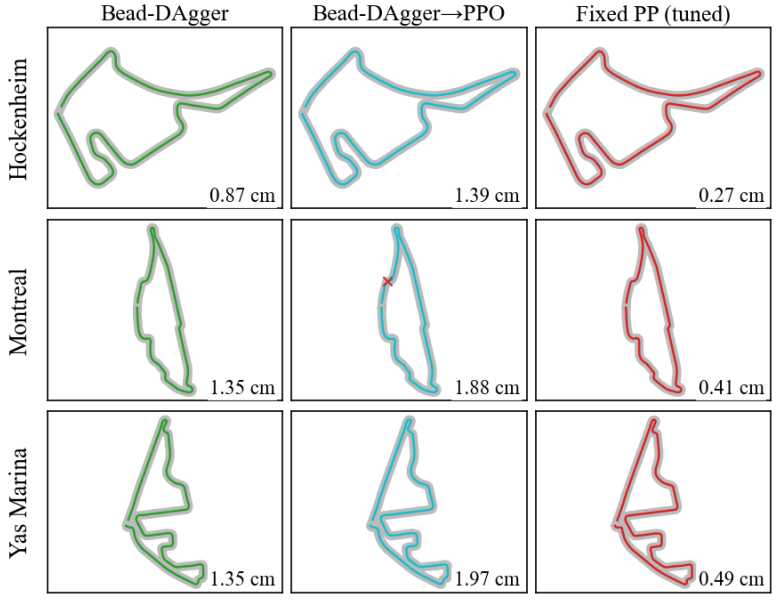


Figure S3: Complete laps (first episode, seed 0) drawn over the grey reference on three unseen circuits. A red cross marks a reference point missed by more than 0.10 m (the start/finish join is ignored); the number in each panel is the RMSE of that lap.

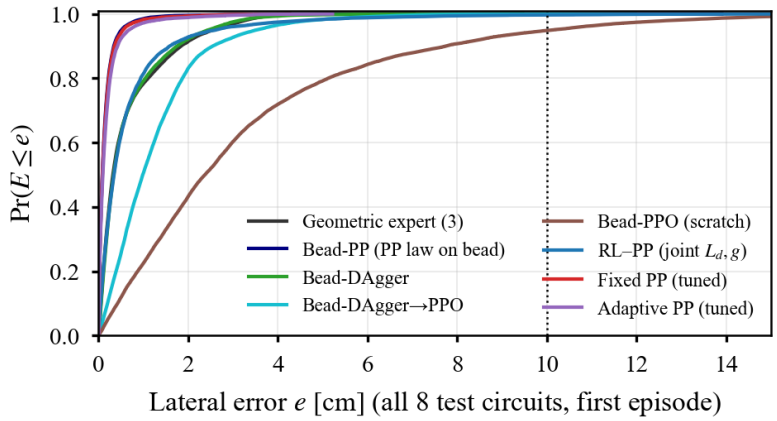


Figure S4: Fraction of trajectory samples whose distance to the extracted reference stays below a threshold, with all eight test circuits taken together (first episode; seed 0 for learned controllers). The dotted line is at 0.10 m.

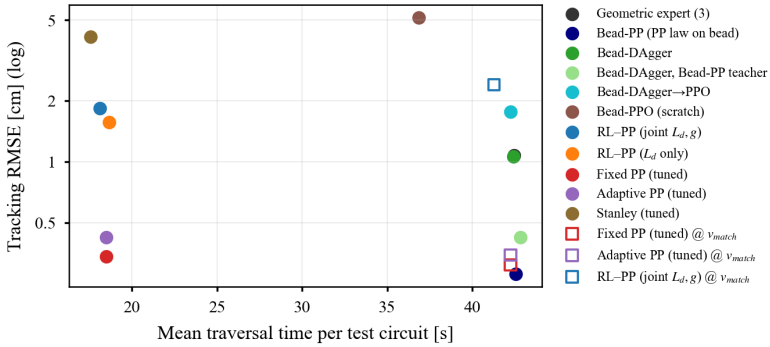


Figure S5: Accuracy against lap time on the test circuits. Solid circles: every controller at its own speed (Track A). Hollow squares: the car's PP-family controllers with their target speed limited to v_{match} (Track B).

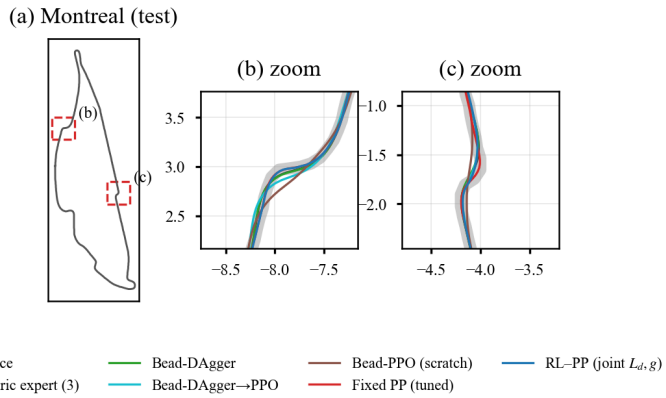


Figure S6: Close-up of the controllers on the unseen Montreal circuit (first episode, seed 0). Both zoom windows are placed by the script at the two reference points where Bead-DAgger→PPO strays furthest, leaving out the start/finish area.